



BEST PRACTICE SERIES

Fluid Analysis Sampling Cart

Application Maintenance Site Management Component Rebuild Safety MARC Management

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**January 15
1206-2.15-1047**

1.0 Introduction

Preventive Maintenance service provides a unique opportunity to plan and execute key Condition Monitoring routines that must be performed at regular intervals. The machine is stopped every 250 or 500 hrs, washed, and remains in the shop for 8 to 12 hours. The planning and scheduling department includes the Condition Monitoring activities in the PM services. The Condition Monitoring, or “Detection” routines normally included in the PM stops are:

1. Visual Inspections
 - a. Pre-Wash Inspection
 - b. During PM Inspection
 - c. Post PM Inspection
2. Performance Tests (including operational tests such as brakes, steering, etc.)
3. Fluid Analysis
 - a. SOS sampling
 - b. Fluids cleanliness (ISO rating)
 - c. Magnetic Plugs inspection
 - d. Magnetic Screens inspection
 - e. Filter cutting / inspection

This Best Practice deals with the handling, organization, and interpretation of the Fluid Analysis samples taken during PM service.

2.0 Best Practice Description

This dealer's facility designed, fabricated and incorporated four (4) small carts to assist in handling and gathering all Fluid Analysis samples taken during PM service.



Sampling Carts assigned to the PM bay

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General View of the Sampling Cart



This picture shows the content of a Cart prepared to be used in the next PM service

Enough compartments were included to allocate space for all Magnetic Plugs, Magnetic and Regular Screens (such as Transmission Screens, Primary Fuel Screen, Hoist Screens, etc.)



These compartments are designed for the Magnetic Plugs, Screens and Filter samples



The compartments located underneath the main cabinet are designed to hold the SOS samples

Implemented working procedure:

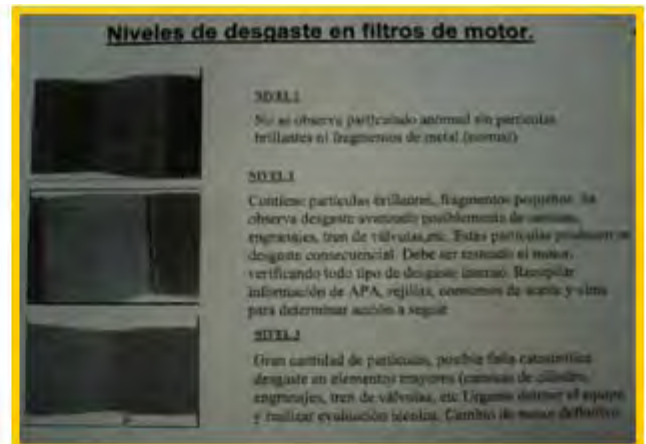
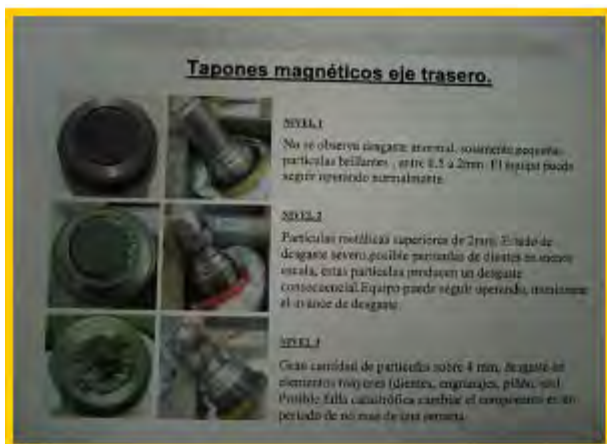
1. Initially these carts are equipped with a set of new Magnetic Plugs and Screens, properly stored in plastic bags.
2. The PM service crew selects a cart and moves it to the specific sampling area at the machine.
3. SOS samples are taken, labeled and stored in the assigned compartment of the cart.

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4. The magnetic plugs are removed, inspected and stored in the compartment of the cart labeled for the specific magnetic plug location. New or recycled magnetic plugs are reinstalled in the machine.
5. The same procedure applies to screens and magnetic screens.
6. The rest of the filters are cut open, inspected, and samples are placed in their respective compartments.
7. Once all samples are gathered, the cart is sent to the technical office.



8. The technical office analyzes the samples, interprets and records the findings. To support the analysis the on-site lab maintains updated guidelines regarding the amount and characteristics of the metallic debris.



Examples of Interpretation Guides for Magnetic Plugs and Engine Oil Filters

9. Pictures are taken for inclusion in the individual electronic files for each machine.

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10. SOS samples are registered in the on-site control system and sent to the local SOS Lab.
11. After the analysis of all samples is completed, the cart with all the reviewed elements is returned to the PM Bay where the parts are recycled and stored in a clean cart, ready for the next service.
12. Damaged parts /elements are replaced with new ones.



The cart filled with recycled materials is made available for the next service.

These carts are currently used to support the sampling routines included in the PM services for a fleet of 793 B and 793C Off-Highway Trucks.

3.0 Implementation Steps

1. Determine the type and quantity of samples to handle based on machine models to be serviced. Consider the number of machines, frequency of service operations, sampling locations (pre-PM and/or PM bay), turnaround time of sampling table (time in the lab to analyze and release the cart), etc.
2. Adopt the design provided in the attachment section of this Best Practice, or customize it to suit particular needs.
3. Fabricate the appropriate number of carts. Stainless steel is the preferred material.
4. Train the PM personnel in the correct procedures.
5. Assign and train a person responsible for receiving and analyzing the samples.

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6. Acquire and equip carts with an exchange set of magnetic plugs / magnetic screens.
7. Assign carts to PM bays and/or pre-PM locations.
8. Implement and control the correct execution of the working procedures.
9. Observe the execution of the implemented process, identify areas for correction, and implement improvements.

4.0 Benefits

The main benefits identified are:

- Improved discipline of Fluid Analysis sampling.
- Minimizes possibility of error from sample mishandling.
- Improves organization of PM routines - resources are allocated in advance.
- Reduces the execution time for sampling activities, which reduces overall PM execution time.
- Improves interpretation accuracy by gathering and providing all samples to one dedicated expert.
- Improved Contamination Control of Fluid Samples.

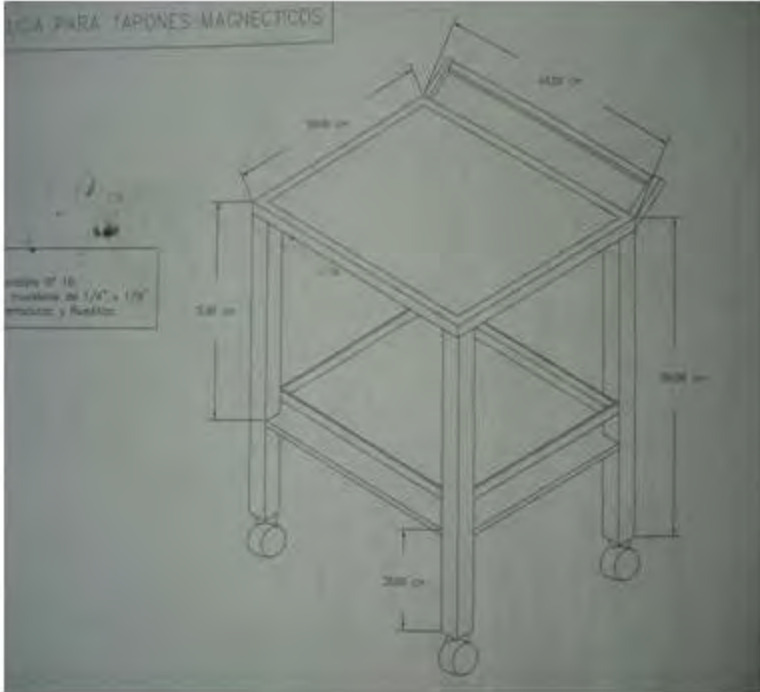
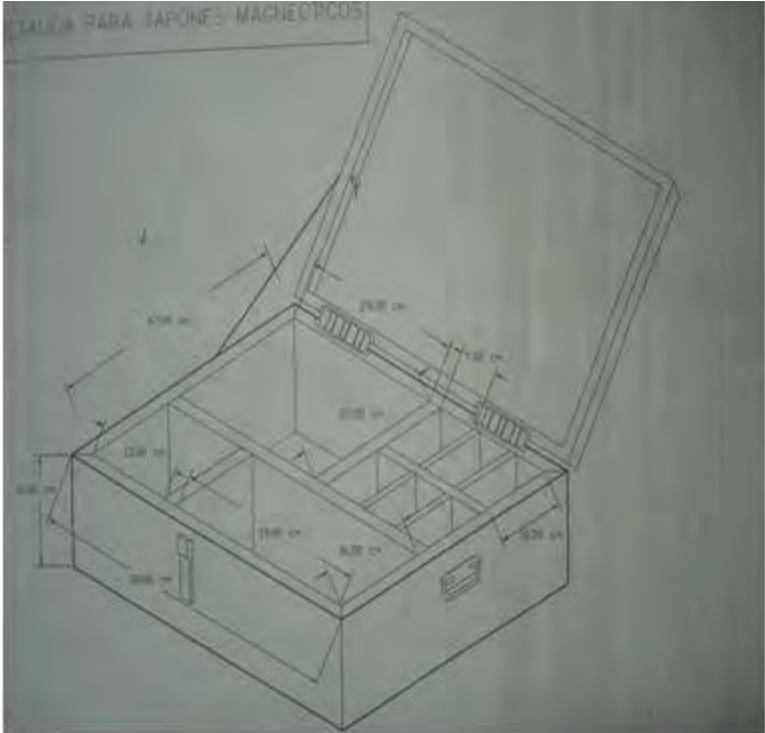
5.0 Resources Required

- Fabricated Cart: the cost of fabricating each cart locally was US \$580. Use this cost as reference only. The number of carts required depends on fleet size and the frequency of the PM services/sampling.
- One set of Magnetic Plugs, Screens, Magnetic Screens per service. The quantity of items and parts number will depend on the specific equipment to service.
- Assign and train one person in the analysis and interpretation of samples that are gathered during PM.

6.0 Supporting Attachments / References

Fabrication Drawings, see below.

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Sampling Cart Drawings

Note: Remember to review and customize the design to your particular needs.

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7.0 Related Best Practices

None at this time.

8.0 Acknowledgements

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For further details and information regarding this Best Practice, please contact the author.

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